Exercise sheet 9 for Numerical Optimization

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Exercise 1 (**Constrained optimization**, 1+2+2 points)

We will now deal with constrained optimization problems. Let $D \subset \mathbb{R}^n$ be open and as well the objective function

$$f : D \subset \mathbb{R}^n \to \mathbb{R}$$

as the constraints

$$g_1, \ldots, g_m,\, h_1, \ldots, h_p : D \to \mathbb{R}$$

be continuously differentiable. We define corresponding index sets

$$\mathcal{I} := \{1, \ldots, m\} \quad \text{and} \quad \mathcal{E} := \{1, \ldots, p\}.$$

The constrained optimization problem reads

$$\min_{x \in D} f(x) \quad \text{such that} \quad g_i(x) \leq 0 \ \forall i \in \mathcal{I} \quad \text{and} \quad h_j(x) = 0 \ \forall j \in \mathcal{E}\ . \tag{1}$$

So we try to minimize f under the constraint that some equalities and inequalities are fulfilled. It can also be just equalities or inequalities, so $\mathcal{I}$ and $\mathcal{E}$ are allowed to be empty. If both are empty, we end up at unconstrained optimization.
The set of all points satisfying the constraints is called *feasible set*,

$$\mathcal{F} := \{x \in D \,|\, g_i(x) \leq 0 \ \forall i \in \mathcal{I},\ h_j(x) = 0 \ \forall j \in \mathcal{E}\}\ .$$

With that, problem (1) becomes

$$\min_{x \in \mathcal{F}} f(x)\ .$$

Consider the exemplary constrained optimization problem

$$\min_{x \in \mathbb{R}^2} x_1 + x_2 \quad \text{such that} \quad x_1^2 \leq 1 - x_2^2 \quad \text{and} \quad x_1 = x_2\ . \tag{2}$$

a) Indicate the objective function and the constraint functions such that (2) takes the form of (1). Also give the corresponding index sets.

b) Draw the subset of $\mathbb{R}^2$ that fulfils the first constraint, the one that fulfils the second constraint and the feasible set $\mathcal{F}$.

c) There is a unique minimizer of the exemplary problem (2). What is this unique minimizer and what is the corresponding objective function value?

Exercise 2 (**Derivation of necessary optimality conditions (KKT conditions)**, 3+3+4+3+3+4+1+1+2+2+2+2+2+3+3+4+3 points)

Our goal is to find computable optimality conditions for the constrained optimization problem (1). For that, we first have to clarify how one can move inside the feasible set. For unconstrained optimization, we could move everywhere inside the domain of f. But now, the possibilities to move are restricted.

Definition 1 (Feasible direction) Suppose a point $x_0 \in \mathcal{F}$. A direction $d \in \mathbb{R}^n$ is called *feasible* if

$$\exists \delta > 0 \quad \text{such that} \quad x_0 + td \in \mathcal{F} \quad \forall t \in [0, \delta] \ ,$$

meaning that one can move along d for a certain distance without leaving $\mathcal{F}$.

We are interested in the set of all feasible directions $C_{fd}(x_0)$.

a) Show that $C_{fd}(x_0)$ is a cone.

What is the connection between the feasibility of a direction and the constraints?

- The index set
$$\mathcal{A}(x_0) := \{i \in \mathcal{I} \,|\, g_i(x_0) = 0\}$$
describes those inequality constraints that become equalities at x_0. They are called *active*. These constraints are critical as there might be directions $d \in \mathbb{R}^n$ with $g_i(x_0 + td) > 0$ $\forall t > 0$.
- The *inactive* inequality constraints, indicated by the index set
$$\mathcal{I} \setminus \mathcal{A}(x_0) = \{i \in \mathcal{I} \,|\, g_i(x_0) < 0\} \ ,$$
are not important as due to continuity for every direction $d \in \mathbb{R}^n$ there exists $s > 0$ such that $g_i(x_0 + sd) < 0$.
- The equality constraints are critical as there might be directions $d \in \mathbb{R}^n$ with $h_j(x_0+td) \neq 0$ $\forall t > 0$.

Consequently, only the active inequality and the equality constraints are relevant for the feasibility of directions.

b) Consider the constrained optimization problem (2) from exercise 1. Is the inequality constraint active at the point $p_0 = (0, 0)$? And at $p_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$? Based on your drawing of the feasible set from exercise 1b), determine $C_{fd}(p_0)$ and $C_{fd}(p_1)$. Make yourself clear what influence the inequality constraint has on the set of feasible directions, depending on whether it is active or not.

Definition 2 (Cone of the descent directions) The *cone of the descent directions* at x_0 is

$$C_{dd}(x_0) := \left\{d \in \mathbb{R}^n \,|\, \nabla f(x_0)^T d < 0\right\} \ .$$

Note: On exercise sheet 8 we defined a cone as a set K with $\alpha x \in K$ for every $x \in K$ and $\alpha \geq 0$. According to this definition, every non-empty cone must contain the zero vector $\mathbf{0}$. But $C_{dd}(x_0)$ cannot contain the zero vector and therefore cannot be a cone. If one slightly changes the definition of a cone to a set K with $\alpha x \in K$ for every $x \in K$ and $\alpha > 0$, $C_{dd}(x_0)$ is a cone. That is how cones are defined in the lecture notes. And therefore I will also call $C_{dd}(x_0)$ a cone.

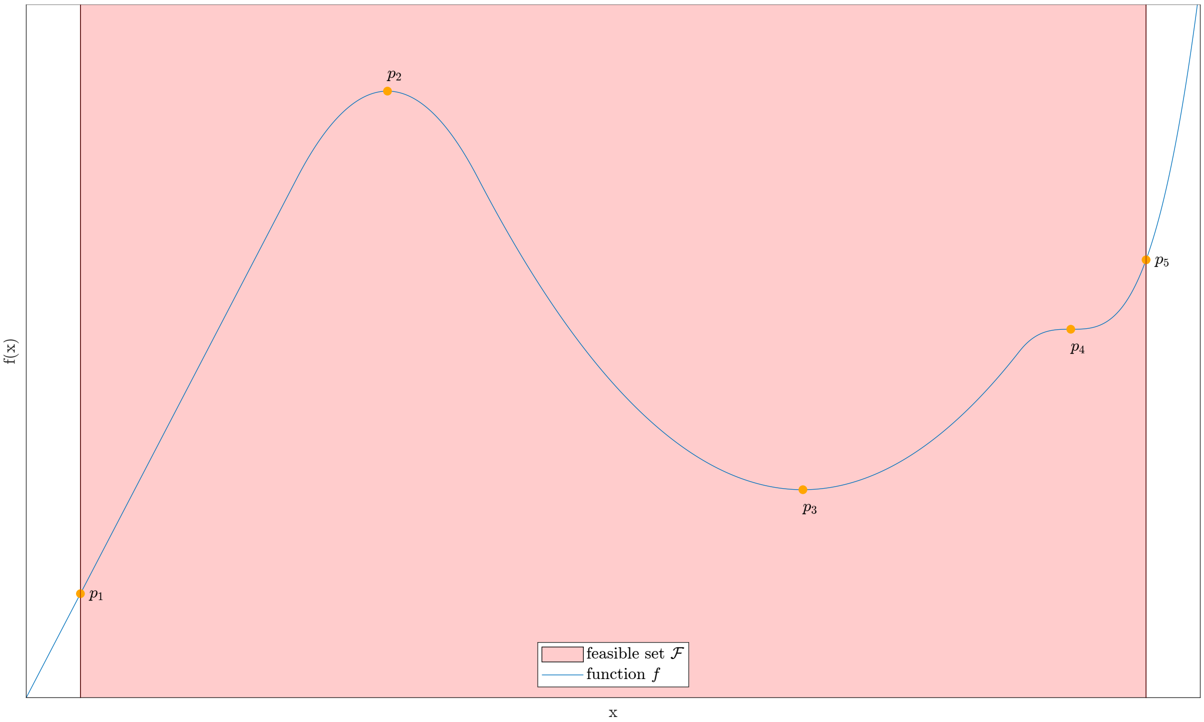

Figure 1: A function $f : \mathbb{R} \to \mathbb{R}$ and a feasible set $\mathcal{F} \subset \mathbb{R}$.

c) Prove that if $x^\star$ is a local minimizer of (1), then

$$C_{dd}(x^\star) \cap C_{fd}(x^\star) = \emptyset \ . \tag{3}$$

d) What does condition (3) mean in simple words? To do this, consider the orange marked points in the plot in figure 1. Why is this condition necessary for a local minimizer? And why not sufficient?

Summarized, condition (3) is a necessary condition for a local minimizer of the constrained optimization problem, but not a sufficient condition.
One might think we have achieved our aim, as we have a necessary condition for a local minimizer. Unfortunately, in general it is not possible to compute the set of feasible directions $C_{fd}(x_0)$. And an optimality condition that cannot be determined does not get us anywhere.

To proceed, we consider a superset of the set of feasible directions, the *linearizing cone* $C_l(g, h, x_0)$.

Definition 3 (Linearizing cone) For $x_0 \in \mathcal{F}$, the *linearizing cone* is defined as

$$C_l(g, h, x_0) := \left\{ d \in \mathbb{R}^n \,|\, \nabla g_i(x_0)^T d \leq 0 \ \forall i \in \mathcal{A}(x_0), \ \nabla h_j(x_0)^T d = 0 \ \forall j \in \mathcal{E} \right\} \ .$$

e) What is the linearizing cone in simple words? To do this, consider the points in the plots in figure 2. What is the linearizing cone in each case? Why must every feasible direction lie in the linearizing cone?

f) Show that for $x_0 \in \mathcal{F}$ it is

$$C_{fd}(x_0) \subset C_l(g, h, x_0) \ . \tag{4}$$

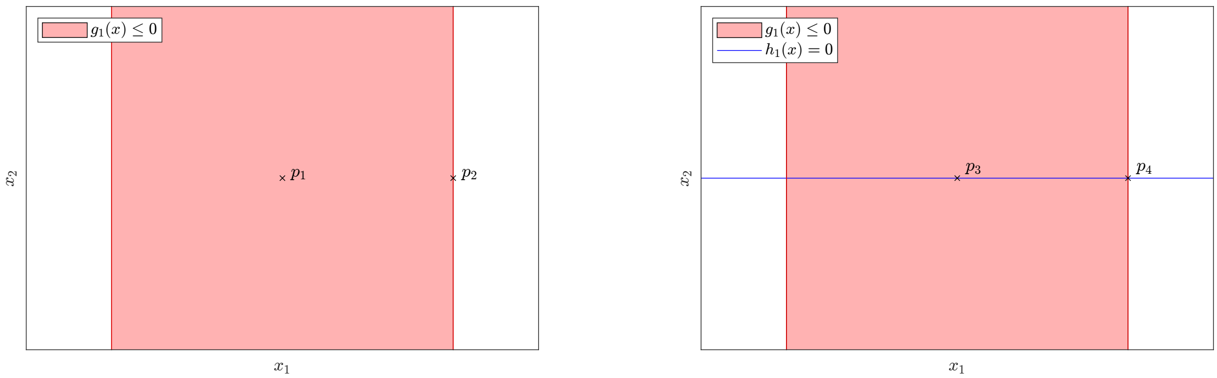

Figure 2: Two feasible sets $\mathcal{F} \subset \mathbb{R}^2$.

Due to (4), the condition

$$C_{dd}(x^\star) \cap C_l(g, h, x^\star) = \emptyset \tag{5}$$

implies (3). But that does not mean that condition (5) is necessary for a local minimizer.

g) With

$$f(x) = (x_1 - 2)^2 + x_2^2\,, \quad g_1(x) = (x_1 - 1)^3 + x_2\,, \quad g_2(x) = -x_2 \quad \text{and} \quad g_3(x) = x_1 - 1$$

consider the constrained problems

$$\min_{x \in \mathbb{R}^2} f(x) \quad \text{such that} \quad g_1(x) \leq 0 \quad \text{and} \quad g_2(x) \leq 0 \tag{6}$$

and

$$\min_{x \in \mathbb{R}^2} f(x) \quad \text{such that} \quad g_1(x) \leq 0\,, \quad g_2(x) \leq 0 \quad \text{and} \quad g_3(x) \leq 0\,. \tag{7}$$

i) Draw the feasible sets of both problems.
ii) Why do both problems have a unique minimizer and what are the respective minimizers?
iii) At the respective minimizers, what are the cones of descent directions?
iv) At the respective minimizers, what are the cones of feasible directions?
v) Draw the cones of descent directions and the cones of feasible directions at the respective minimizers. Is condition (3) fulfilled at the respective minimizers and why does it have to be?
vi) At the respective minimizers, what are the linearizing cones? Draw them.
vii) Is condition (5) fulfilled at the respective minimizers? What is the reason for that condition (5) is fulfilled for one problem but not for the other, and what are no reasons? What does that mean for condition (5) being necessary for a local minimizer?

Condition (5)'s right to exist is that it is equivalent to a condition that is easy to compute, the so-called *KKT conditions*.

Definition 4 (KKT conditions) The *KKT conditions* assume that for $x_0 \in \mathcal{F}$ there exist so-called *Lagrange multipliers* $\lambda \in \mathbb{R}^m_{\geq 0}$ and $\mu \in \mathbb{R}^p$ such that

$$\nabla f(x_0) + \sum_{i=1}^{m} \lambda_i \nabla g_i(x_0) + \sum_{j=1}^{p} \mu_j \nabla h_j(x_0) = \mathbf{0}$$

and

$$\lambda_i g_i(x_0) = 0 \;\forall i \in \mathcal{I}\,.$$

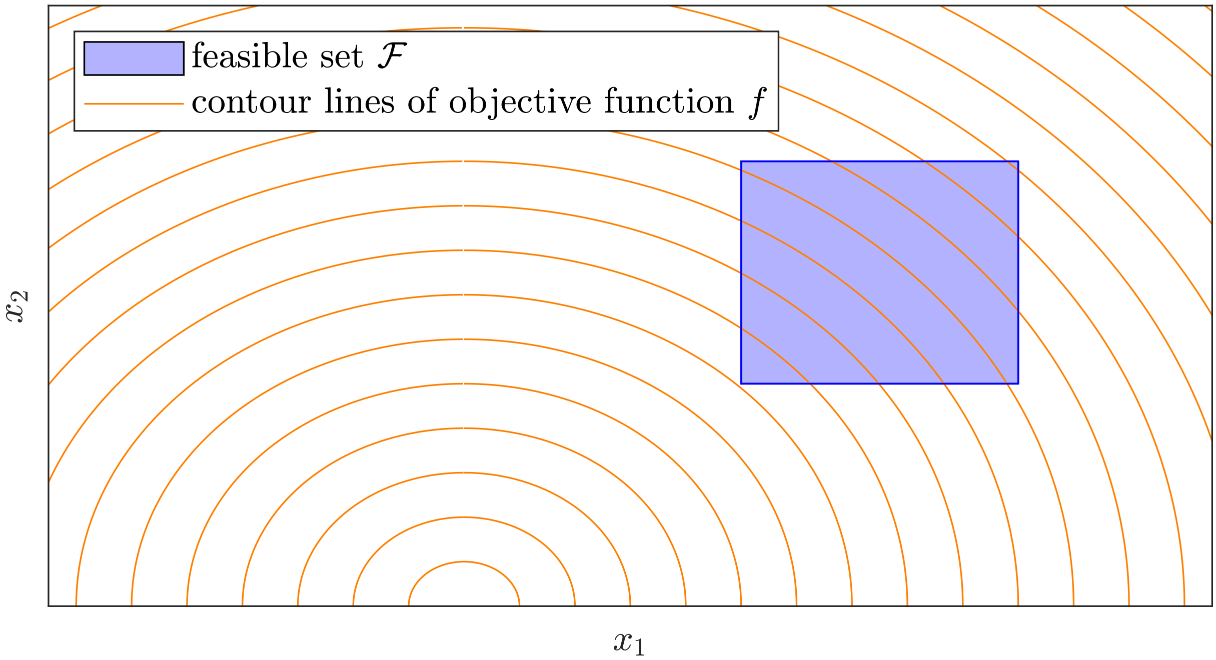

Figure 3: Sketch of an optimization problem in $\mathbb{R}^2$. The objective function f is supposed to be an upward-opened parabola.

h) What do the KKT conditions say in simple words? Especially about linear dependence.

i) Consider figure 3. There are four points in the feasible set, that are candidates for being a KKT point. Which are they and why?
Actually, there is only one KKT point. Which is it and why?

The KKT conditions being fulfilled is equivalent to condition (5).

j) For simplicity's sake, we only want to prove one implication of the equivalence. Prove that if the KKT conditions are fulfilled at $x^\star \in \mathcal{F}$, then condition (5) is satisfied.

k) For both example problems from exercise g), check if the KKT conditions are satisfied at the respective unique minimizers. If so, compute the Lagrange multipliers. Are they unique? Verify that your results comply with condition (5).

That's it for this sheet. We have the implications marked in black in the following logical chain:

$$
\begin{array}{ccc}
x^\star \text{ local minimizer} & \Longrightarrow & C_{dd}(x^\star) \cap C_{fd}(x^\star) = \emptyset \\
& & \Uparrow \; \color{red}{\Downarrow} \\
& & C_{dd}(x^\star) \cap C_l(g,h,x^\star) = \emptyset \iff \text{KKT holds}
\end{array}
$$

With that, the KKT conditions are not necessary for a local minimizer yet. On the next sheet, we will take care of the implication marked in red and obtain necessity. However, the implication marked in red will not apply to every optimization problem, but only to those that satisfy so-called *constraint qualifications*. You have already seen the consequences of this in exercise g), where one of two seemingly identical optimization problems was formulated unfavourably and therefore did not satisfy the KKT conditions at a local minimizer.